Aashray Jain

Full-Stack & AI/ML Engineer | M.Tech Candidate in AI & ML

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To: The Hiring Team / Engineering Selection Committee
Focus Area: Artificial Intelligence, Machine Learning, Generative AI & Agentic Systems
Subject: Application for Opportunities in AI / ML, Agentic Systems & Software Engineering

Dear Hiring Team,

I am writing to express my strong interest in joining your organization in an **Artificial Intelligence, Machine Learning, or Agentic Systems** engineering role. As an engineer with over four years of professional software experience and a current **M.Tech candidate in AI & Machine Learning** at NIMS Marik Institute of Computing, AI, Robotics & Cybernetics, NIMS University Rajasthan (holding a B.Tech with an **8.74 / 10.0 CGPA Distinction**), I bring a proven track record of bridging advanced machine learning formulations with high-throughput, production-grade distributed architectures.

Throughout my career, I have focused on solving real-world challenges at the intersection of foundation models, algorithmic rigor, and dependable systems engineering:

- **Production Agentic Systems & Protocol Architecture (Novafy OU):** As a core contributor to the ground-up rebuild of NovaAgent v2 in Python and FastAPI (authoring 190+ reviewed commits), I architected composable tools consuming and serving interfaces over the **Model Context Protocol (MCP)**. I designed a *Human-in-the-Loop formal safety verification layer* for enterprise ERP (Microsoft Dynamics 365 Business Central) that enforces pre-execution live metadata validation and non-hallucinatory value substitution. Additionally, I implemented request restructuring protocols for provider-side prompt caching achieving **∼99% cached tokens**, conversation compaction algorithms, and an automated evaluation harness with 70+ deterministic scenarios to quantify reasoning drift.
- **Deep Learning Research & Explainable AI (ArihantQNet):** I designed and evaluated an end-to-end deep learning framework for multi-stage Alzheimer's disease stratification across 6,400 T1-weighted axial brain MRI scans under a deterministic 70/15/15 stratified partition. To overcome severe 49.78:1 class skew, I derived a *Normalized Square-Root Inverse-Frequency Weighted Cross-Entropy Loss* coupled with AdamW, attaining 100% sensitivity on rare pathologies. I formulated **ArihantQNet**, a novel tri-branch hybrid network integrating ResNet50, DenseNet121, and EfficientNet-B0 via Quantum-Inspired Multi-Head Cross-Attention (Q-MCA), and validated clinical decision boundaries using Grad-CAM neuroanatomical heatmaps.
- **Retrieval-Augmented Generation & Multi-Modal AI:** I engineered high-performance RAG pipelines using FAISS, LangChain, and open-weights models (DeepSeek, Mixtral, Zephyr) with optimized semantic chunking to eliminate hallucination. My applied AI skills are backed by Microsoft Applied Skills certifications in Generative AI (Azure OpenAI) and Document Intelligence, alongside Google Cloud credentials in Multimodal RAG and Vertex AI.
- **Distributed Cloud Backends & Applied AI (TCS & ArihantQ):** Over 3.5 years as a Systems Engineer at Tata Consultancy Services, I engineered high-throughput, real-time risk platforms for Canadian BFSI serving 500K+ enterprise users, spearheaded multi-account AWS migrations that reduced infrastructure costs by 70%, and collaborated with TCS Rapid Labs on applied AI/ML proof-of-concepts. Through my independent lab ArihantQ, I won the Lyzr AI Architect Challenge for algorithmic LLM cost optimization (projecting 666% ROI) and received public commendation from the MariaDB Foundation & CEO for open-source database innovation.

What distinguishes my profile is my end-to-end capability: I am equally at home deriving custom loss functions and attention mechanics in PyTorch as I am designing resilient REST/MCP microservices, optimizing AWS cloud infrastructure, or building human-in-the-loop validation pipelines. I thrive in collaborative, fast-moving environments that value empirical verification, clean code, and user-centric impact.

I would welcome the opportunity to discuss how my technical expertise in Generative AI, agentic protocols, and production systems engineering can contribute to your team's upcoming initiatives. Thank you for your time and consideration.

Sincerely,

Aashray Jain
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